

RAVIVARMA PATTURI

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Professional Profile

AI/ML Engineer | 4+ Years | Orchestrating solutions for leading U.S. healthcare payers and providers | Product & AI Companies | Generative AI, LLMs, Agentic AI, RAG, NLP, Multi Agent Systems | PyTorch, Hugging Face, LangChain, FastAPI

AI/ML Engineer with 4+ years of experience designing and deploying production-grade Generative AI, Agentic AI, and NLP systems — from LLM-driven multi-agent platforms and RAG pipelines to deep learning models — for leading U.S. healthcare payers, providers, and product-focused AI companies.

Work Experience

AI Engineer, Autonomize AI

Feb 2026 - Present

Genesis AI Studio – AI Platform for Building & Deploying Agents

- Designed and implemented the **Recursive Language Model (RLM) architecture**, enabling **hierarchical reasoning** through **Master-Sub LLM collaboration**.
- Researched and integrated the RLM strategy into the existing agent platform alongside **ReAct** and **Plan-and-Execute** workflows.
- Built the orchestration workflow using **Temporal**, enabling **scalable execution** of production-grade AI agents.

Healthcare AI – Benefits Digitization

- Successfully delivered** a proof of concept for **benefits digitization** for a leading U.S. healthcare organization within a **14-day timeline**, with its success leading to **conversion into a full-scale production project**.
- Built AI pipelines to extract structured benefit information from policy documents using **LLMs and Knowledge Graphs**.
- Gained expertise in U.S. healthcare standards, including **ICD and CPT coding systems**.

Clinical AI Expression & Semantic Search

- Built an **NLP pipeline** to convert natural language business requests into structured logical expressions for campaign creation.
- Indexed more than **70,000 healthcare concepts** using **Azure AI Search** and improved retrieval quality using **Cross-Encoder reranking**.
- Automated expression generation, significantly **reducing manual effort** in campaign creation.

SOP Reasoner for Automated Claims Adjudication (Multi-Agent System)

- Developed a **multi-agent AI system** that automates healthcare claims adjudication by reasoning over **Standard Operating Procedure (SOP)** documents, **reducing adjudication cost by 80%** compared to the existing solution.
- Built a **Knowledge Graph Generation Agent** that converts SOPs into a structured decision graph using **conditional (if-else) business rules** for explainable reasoning.
- Implemented an **Adjudication Agent** that processes claim inputs, traverses the knowledge graph, evaluates policy rules step-by-step, and determines the final adjudication decision.
- Generated **explainable outputs** with the final claim decision (**Approve/Reject/Pend**) and detailed step-by-step reasoning, improving **transparency and auditability**.

Healthcare Inquiry Classification

- Developed a **ModernBERT-based** text classification model for healthcare inquiry categorization.
- Addressed **class imbalance** through **synthetic data generation** and **oversampling techniques**, improving classification performance.
- Evaluated the model using **precision, recall, F1-score, and confusion matrix** to ensure robust production performance.

AI Research Engineer, Boltzmann Labs Pvt Ltd

July 2023 - Jan 2026

AI Modelling of Battery Management System

- Implemented sequential neural networks (RNN, LSTM, GRU) using PyTorch Framework, achieving 99% reliability in SOC predictions. and Later **re-implemented the same models from scratch**.
- Applied **dynamic quantization** and inference optimization, reducing model size by **50%** and improving runtime latency by **2×**, with **<1.2% deviation** in MAPE — enabling deployment on **resource-constrained edge devices** for real-time BMS monitoring.

- Designed a **novel surrogate model** by incorporating physics into neural networks, enhancing model interpretability and insights into battery behavior.

Automated LLM Driven Agentic Workflow for RTL Verification

- Developed an LLM-powered agentic verification workflow** to automatically extract functional intent from IC specifications, enabling precise, coverage-aligned test-plan generation with targeted stimulus.
- Integrated the AI agent into Cadence Xcelium UVM flows**, reducing manual verification effort and **accelerating coverage closure** through intelligent, automated test strategy formulation.
- Experimented with both open-source and proprietary LLMs** to optimize task success rates, model reasoning performance, and adaptability to complex semiconductor design specifications.

Molecular Property Prediction

- Built a robust end-to-end ML pipeline** covering data preprocessing, feature selection, model evaluation, and **hyperparameter optimization**, ensuring reproducible and generalizable performance across diverse chemical structures.
- Built ML & DL models for molecular property prediction, enabling accurate **molecular screening** and **similarity search for candidate selection**.

PoC for Battery Chemistry

- Developed a domain-specific RAG system** using LangChain, Mistral API, and ChromaDB, delivering **accurate, context-aware retrieval** from heterogeneous battery research data to accelerate scientific analysis.
- Integrated and production-tested open-source LLMs within in-house workflows, focusing on deployment, prompt engineering, and debugging across **multi-component generative AI systems** to enhance researcher productivity.
- Fine-tuned LLaMA-3.2-1B using LoRA** on NVIDIA L40S GPUs for **chemically valid crystal structure generation**, enabling unconditional, conditional, and infilling prompts for **novel material discovery** in battery research.

Automation Workflow for Gromacs

- Engineered a GPU-accelerated and Dockerized workflow** for Molecular Dynamics (MD) simulations using GROMACS + OpenMM, reducing environment setup time by **40%**.
- Automated protein-ligand preparation and production pipelines, boosting workflow efficiency by **30%** and ensuring reproducible simulation runs.
- Deployed the fully **containerized** MD automation pipeline as a **FastAPI microservice** on **AWS Kubernetes (EKS)** using Docker images, enabling scalable GPU job execution and reproducible cloud-based simulations.

Deep Learning Research Intern, Boltzmann Labs Nov 2022 - Jun 2023

- Contributed to Boltchem for small molecule drug discovery, building robust ML models.
- Researched on Multi-objective optimization to improve multiple properties simultaneously and implemented them.

Technical Skills

- Programming skills & Databases:** Python, C , C++ , MySQL
- Libraries:** Pandas, Numpy, Matplotlib, Pytorch, Tensorflow, HuggingFace, NLTK, SpaCy.
- Deployment:** Docker, Linux , Fastapi, AWS, Streamlit, FastAPI, Git , Github , Kubernetes
- Machine & Deep Learning:** Regression , Classification , Clustering, ANNs, LSTMs, RNNs, CNNs , Transformers , AutoEncoders
- Gen AI:** LangChain, LLamaIndex, LLMs , Prompt Engineering , RAG , Agents , LangGraph, Finetuning, Temporal, RLM Architectures

Education & Certification

Rajiv Gandhi University of Knowledge and Technologies (B-Tech,ECE)	CGPA : 9.0	Aug 2019 - May 2023
Analytics Vidhya, Gen AI Pinnacle Program		May 2024 - Dec 2025
Neural Networks & Deep Learning ,Coursera		
PyTorch for Deep Learning with Python Bootcamp , Udemy		

Projects

Regulatory QA RAG Chatbot | [Github Link](#) | [Streamlit](#)

- Built a user-friendly QA RAG chat system** using OpenAI APIs (GPT-4o) and Streamlit, designed for accurate, context-aware responses and smooth conversational interaction for end-users.

- **Optimized the full RAG pipeline** with **flexible chunking**, **robust document parsing**, **hybrid retrieval**, **query rewriting**, and **memory-augmented context** handling — improving relevance, grounding, and performance.
- **Built and deployed the solution end-to-end** by containerizing the backend and frontend with Docker and deploying on **Kubernetes**, enabling scalable access and reliable performance in production environments.

AI-Powered Financial Performance Reporting Agent | [Link](#)

- **Built an AI-driven financial reporting agent** using LangGraph and LangChain workflows integrated with OpenAI GPT models to analyze company financial data and generate accurate, narrative-rich performance insights.
- Enabled **automated competitor benchmarking** by integrating Tavily Search API to retrieve and compare key financial metrics, supporting more informed strategic decision-making.
- Developed an interactive Streamlit UI supporting CSV uploads, iterative refinement, and feedback-driven report enhancement — improving accessibility and usability for finance teams and decision-makers.

Traffic Volume Prediction Model Leveraging Advanced ML Techniques | [Link](#)

- Developed a high-performance traffic volume prediction model leveraging XGBoost with advanced feature engineering techniques for accurate forecasting.
- Implemented context-aware feature engineering, incorporating weather, time, and holiday features to ensure robust and reliable predictions.
- Achieved a **Top 10 Rank in the MachineHack Great Indian Hackathon 2024**

Emotion Detection Using Machine Learning on GoEmotions Corpus | [Link](#)

- Built a **multi-class emotion classifier** using the GoEmotions dataset by grouping 27 emotion labels into 5 major categories.
- Applied standard NLP preprocessing and feature extraction techniques such as **TF-IDF** to prepare text for modeling.
- Trained and evaluated multiple models including **Logistic Regression**, **Naive Bayes**, and **Neural Networks** to identify the most reliable approach for emotion classification.

Learning & Community Engagement:

- (<https://www.youtube.com/@neuralmastery>) - always eager to learn , love to share knowledge and make meaningful contributions to Tech.
- **Deep-ML**: <https://www.deep-ml.com/profile/uejNhzY3qPWAfsAXPmXvt6tXDp1> (Current Rank : **TOP 100**)
- **Hackerrank**: <https://www.hackerrank.com/profile/ravivarmapatturi>
- **Leetcode**: <https://leetcode.com/u/ravivarmapatturi/>
- Currently learning to **build multi-agent systems** using LLMs to enable collaborative problem-solving, autonomous decision-making, and complex task coordination.
- Keep exploring **advanced reinforcement learning** techniques such as policy gradients, actor-critic methods, and deep Q-networks to build intelligent decision-making systems.
- Reading latest Research Papers in AL/ML